

# **Introduction to Lie Group Theory**

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Summer 2026

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# Chapter 1

## Matrix Groups

### 1.1 Groups of Matrices

Let  $M_{m,n}(\mathbb{K})$  be the set of  $m \times n$  matrices whose entries are in  $\mathbb{K}$ , a commutative field. We will also use the special notations  $M_n(\mathbb{K}) = M_{n,n}(\mathbb{K})$ , and  $\mathbb{K}^n = M_{n,1}(\mathbb{K})$ .

$M_{m,n}(\mathbb{K})$  is a vector space over  $\mathbb{K}$  with dimension  $mn$ . As well as being a vector space over  $\mathbb{K}$  with dimension  $n^2$ ,  $M_n(\mathbb{K})$  is also a ring with the usual addition and multiplication of matrices. It is also an important example of a finite dimensional algebra over  $\mathbb{K}$ .

We will use the notation

$$GL_n(\mathbb{K}) = \{A \in M_n(\mathbb{K}) : \det A \neq 0\}$$

for the set of invertible  $n \times n$  matrices, known as the  $n \times n$  *general linear group*, and

$$SL_n(\mathbb{K}) = \{A \in M_n(\mathbb{K}) : \det A = 1\} \subseteq GL_n(\mathbb{K})$$

for the set of  $n \times n$  unimodular matrices, known as the  $n \times n$  *special linear group*.

**Theorem 1.1.** *The sets  $GL_n(\mathbb{K})$  and  $SL_n(\mathbb{K})$  are groups under matrix multiplication. Furthermore,  $SL_n(\mathbb{K})$  is a subgroup of  $GL_n(\mathbb{K})$ .*

Since determinant of a matrix is a polynomial in the entries of the matrix, it can be proved that the determinant function

$$\det : M_n(\mathbb{K}) \longrightarrow \mathbb{K}$$

is continuous.

**Proposition 1.1.** Let  $\mathbb{K} = \mathbb{R}$  or  $\mathbb{C}$ . Then

- (i)  $GL_n(\mathbb{K}) \subseteq M_n(\mathbb{K})$  is an open subset;
- (ii)  $SL_n(\mathbb{K}) \subseteq M_n(\mathbb{K})$  is a closed subset.

*Proof.*  $GL_n(\mathbb{K}) = M_n(\mathbb{K}) - \det^{-1}\{0\}$ , which is open since  $\{0\}$  is closed. Similarly,  $SL_n(\mathbb{K}) = \det^{-1}\{1\}$ , which is closed since the singleton  $\{1\}$  is closed in  $\mathbb{K}$ . ■

**Definition 1.1.** Let  $G$  be a topological space, and view  $G \times G$  as the product space. Suppose that  $G$  is also a group with multiplication map  $\text{mult} : G \times G \rightarrow G$ , and inverse map  $\text{inv} : G \rightarrow G$ .  $G$  is a *topological group* if  $\text{mult}$  and  $\text{inv}$  are continuous.

**Theorem 1.2.** For  $\mathbb{K} = \mathbb{R}$  or  $\mathbb{C}$ ,  $GL_n(\mathbb{K})$  and  $SL_n(\mathbb{K})$  are topological groups.

## 1.2 Matrix Groups

**Definition 1.2.** A subgroup  $G$  of  $GL_n(\mathbb{K})$  is known as a *matrix group*, given it is a closed subset of  $GL_n(\mathbb{K})$ .

*Remark.* We may consider  $GL_n(\mathbb{K})$  as a subgroup of  $GL_{n+1}(\mathbb{K})$  by identifying the  $n \times n$  matrix  $A = [a_{ij}]$  with

$$\begin{bmatrix} A & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & \cdots & a_{1n} & 0 \\ \vdots & \ddots & \vdots & \vdots \\ a_{n1} & \cdots & a_{nn} & 0 \\ 0 & \cdots & 0 & 1 \end{bmatrix}$$

And it is easily verified that  $GL_n(\mathbb{K})$  is closed in  $GL_{n+1}(\mathbb{K})$ . With the aid of this embedding, any matrix subgroup of  $GL_n(\mathbb{K})$  can also be viewed as a matrix subgroup of  $GL_{n+1}(\mathbb{K})$ .

**Definition 1.3.** Let  $G, H$  be two matrix groups. A groups homomorphism  $\varphi : G \rightarrow H$  is a *continuous homomorphism of matrix groups* if it is continuous and its image  $\varphi(G)$  is a closed subgroup of  $H$ .

**Examples.** Some important examples of real and complex matrix groups:

### 1. Groups of Upper Triangular Matrices

For  $n \geq 1$ , an  $n \times n$  matrix  $A = [a_{ij}]$  is said to be *upper triangular* if  $a_{ij} = 0$  if  $i < j$ . An upper triangular matrix is said to be *unipotent* if all of its diagonal entries are 1.

The *upper triangular subgroup* or *Borel subgroup* of  $GL_n(\mathbb{K})$  is

$$UT_n(\mathbb{K}) = \{A \in GL_n(\mathbb{K}) : A \text{ is upper triangular}\},$$

while the *unipotent subgroup* of  $GL_n(\mathbb{K})$  is

$$SUT_n(\mathbb{K}) = \{A \in GL_n(\mathbb{K}) : A \text{ is unipotent}\}.$$

It is easy to verify that  $UT_n(\mathbb{K})$  and  $SUT_n(\mathbb{K})$  are closed subgroups of  $GL_n(\mathbb{K})$ , and that  $SUT_n(\mathbb{K})$  is a subgroup of  $UT_n(\mathbb{K})$ .

## 2. Affine Groups

The  $n$ -dimensional *affine group* over  $\mathbb{K}$  is

$$Aff_n(\mathbb{K}) = \left\{ \begin{bmatrix} A & t \\ 0 & 1 \end{bmatrix} : A \in GL_n(\mathbb{K}), t \in \mathbb{K}^n \right\}$$

which is clearly a closed subgroup of  $GL_{n+1}(\mathbb{K})$ . If we identify  $x \in \mathbb{K}^n$  with  $\begin{bmatrix} x \\ 1 \end{bmatrix} \in \mathbb{K}^{n+1}$ , then as a consequence of

$$\begin{bmatrix} A & t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ 1 \end{bmatrix} = \begin{bmatrix} Ax + t \\ 1 \end{bmatrix}$$

we obtain an action of  $Aff_n(\mathbb{K})$  on  $\mathbb{K}^n$ .

The group  $Aff_n(\mathbb{K})$  can be expressed as the semi-direct product of  $Trans_n(\mathbb{K})$  and  $GL_n(\mathbb{K})$ , where

$$Trans_n(\mathbb{K}) = \left\{ \begin{bmatrix} I_n & t \\ 0 & 1 \end{bmatrix} : t \in \mathbb{K}^n \right\}$$

is a closed normal subgroup of  $Aff_n(\mathbb{K})$ . That is,

$$Aff_n(\mathbb{K}) = GL_n(\mathbb{K}) \ltimes Trans_n(\mathbb{K}).$$

## 3. Orthogonal Groups

The  $n \times n$  *real orthogonal group* is the subset

$$O(n) = \{A \in GL_n(\mathbb{R}) : A^T A = I_n\},$$

where  $A^T$  is the transpose of  $A = [a_{ij}]$  whose entries are given by  $(A^T)_{ij} = a_{ji}$ .  $O(n)$  is a closed subgroup of  $GL_n(\mathbb{R})$ . For all  $A \in O(n)$ , since  $\det A = \det A^T$ ,  $\det A = \pm 1$ . The subgroup

$$SO(n) = \{A \in O(n) : \det A = 1\}$$

is known as the *special orthogonal group*. One of the main reasons for the study of orthogonal groups is due to their relation with isometries. If  $A \in GL_n(\mathbb{R})$ , then it can be proved that  $A$  is a linear isometry *iff*  $A$  is orthogonal. The elements of  $SO(n)$  are called *direct isometries* or *rotations*.

#### 4. Unitary Groups

The  $n \times n$  *unitary group* is the subset

$$U(n) = \{A \in GL_n(\mathbb{C}) : A^*A = I_n\},$$

where  $A^*$  is the *hermitian conjugate* of  $A$ , that is  $A^*)_{ij} = a_{ji}$ .  $U(n)$  is clearly a closed subgroup of  $GL_n(\mathbb{C})$ . The *special unitary group* is the subset

$$SU(n) = \{A \in U(n) : \det A = 1\},$$

which is also a normal subgroup of  $U(n)$ .

#### 5. Symplectic Groups

A matrix  $S$  is called *skew symmetric* if  $S^T = -S$ . The standard example of this is built up using the  $2 \times 2$  block

$$J = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

When  $m \geq 1$  we have the non degenerative skew symmetric matrix

$$J_{2m} = \begin{bmatrix} J & O_2 & \cdots & O_2 \\ O_2 & J & \cdots & O_2 \\ \vdots & \vdots & \ddots & \vdots \\ O_2 & O_2 & \cdots & J \end{bmatrix}.$$

The matrix group

$$\text{Symp}_{2m}(\mathbb{R}) = \{A \in GL_{2m}(\mathbb{R}) : A^T J_{2m} A = J_{2m}\}$$

is called the  $2m \times 2m$  *real symplectic group*. It can be proved that if  $A \in \text{Symp}_{2m}(\mathbb{R})$ , then  $\det A = 1$ .

### 1.3 Complex Matrices as Real Matrices

Similar to how complex numbers can be viewed as a 2-dimensional real vector space with basis  $(1, i)$ , every  $n \times n$  complex matrix can also be viewed as a  $2n \times 2n$  real matrix using the following construction or the ones similar to it.

We identify each complex number  $z = x + iy$  with a  $2 \times 2$  matrix by defining a function

$$\rho : \mathbb{C} \longrightarrow M_2(\mathbb{R}); \quad \rho(x + iy) = \begin{bmatrix} x & -y \\ y & x \end{bmatrix}.$$

This can also be expressed as

$$\rho(x + iy) = xI_2 - yJ.$$

As  $\rho$  is an injective ring homomorphism, we can view  $\mathbb{C}$  as a subring of  $M_2(\mathbb{R})$ . Notice that  $\rho(\bar{z}) = \rho(z)^T$ .

For a  $n \times n$  complex matrix  $Z$ , we define a function  $\rho_n : M_n(\mathbb{C}) \longrightarrow M_{2n}(\mathbb{R})$ ;

$$\rho_n(Z) = \begin{bmatrix} \rho(z_{11}) & \rho(z_{12}) & \cdots & \rho(z_{1n}) \\ \rho(z_{21}) & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \vdots \\ \rho(z_{n1}) & \cdots & \cdots & \rho(z_{nn}) \end{bmatrix}$$

Then  $\rho_n$  is a continuous injective ring homomorphism. Generally for  $n \times n$  real matrices  $X$  and  $Y$ ,

$$\rho_n(X + iY) = \rho_n(X) - J_{2n}\rho_n(Y)J_{2n}.$$

### 1.4 Matrix Groups of Normed Vector Spaces

It is often desirable to generalize the notion of matrix spaces to arbitrary finite dimensional real or complex vector spaces. Here we assume  $\mathbb{K} = \mathbb{R}$  or  $\mathbb{C}$ .

**Theorem 1.3.** *Let  $(U, \mu)$  and  $(V, \nu)$  be two finite dimensional normed vector spaces of the same dimension over  $\mathbb{K}$ . Then any linear isomorphism  $\varphi : U \longrightarrow V$  is a homeomorphism.*

**Corollary.** *If  $\mu$  and  $\nu$  are two norms on a finite dimensional vector space, then they give rise to the same topology.*

*Proof.* The identity function is a linear isomorphism, which must be continuous. This implies that every open set in the topology of  $\nu$  is open in the topology of  $\mu$  and vice-versa. ■

Given a finite dimensional normed vector space  $V$ ,  $\nu$  and a linear transformation  $\alpha : V \rightarrow V$ , we define the *operator norm* of  $\alpha$  with respect to  $\nu$  by

$$\|\alpha\|_\nu = \sup\{\nu(\alpha(x)) : x \in V, \nu(x) = 1\}$$

This defines a norm on  $\text{End}_{\mathbb{K}}(V)$ , the set of all linear transformations  $V \rightarrow V$ , which makes it a finite dimensional normed vector space over  $\mathbb{K}$ .

**Proposition 1.2.** *If  $\mu$  and  $\nu$  are two norms on a finite dimensional vector space  $V$ , then the topologies associated to  $\text{End}_{\mathbb{K}}(V)$  with respect to  $\mu$  and  $\nu$  are the same.*

Inside  $\text{End}_{\mathbb{K}}(V)$  we have the group of linear isomorphisms,  $\text{GL}_{\mathbb{K}}(V) \subseteq \text{End}_{\mathbb{K}}(V)$ , the *general linear group* of the vector space  $V$ .

**Proposition 1.3.** *Let  $U$  and  $V$  be two vector spaces of the same dimension. If  $\varphi : U \rightarrow V$  is a linear isomorphism, then it induces a continuous group isomorphism with a continuous inverse,*

$$\varphi_* : \text{GL}_{\mathbb{K}}(U) \rightarrow \text{GL}_{\mathbb{K}}(V); \quad \varphi_*(\alpha) = \varphi \circ \alpha \circ \varphi^{-1}$$

Generally, given a finite basis for  $V$ , there is an associated linear isomorphism  $V \cong \mathbb{K}^n$  for some  $n$ , a ring isomorphism  $\text{End}_{\mathbb{K}}(V) \cong M_n(\mathbb{K})$  and a group isomorphism  $\text{GL}_{\mathbb{K}}(V) \cong \text{GL}_n(\mathbb{K})$ .

**Theorem 1.4.** *Let  $V$  be a finite dimensional vector space over  $\mathbb{K}$  with dimension  $n$  and let  $\varphi : V \rightarrow \mathbb{K}^n$  be a linear isomorphism. Then  $\varphi_* : \text{GL}_{\mathbb{K}}(V) \rightarrow \text{GL}_n(\mathbb{K})$  is a continuous group isomorphism with a continuous inverse. Hence  $\text{GL}_{\mathbb{K}}(V)$  is a matrix group.*

## 1.5 Matrix Exponential and Logarithm

We define the exponential and logarithm functions for matrices using their power series. Throughout this section, we assume  $\mathbb{K} = \mathbb{R}$  or  $\mathbb{C}$ .

The *exponential function* is,  $\exp : M_n(\mathbb{K}) \rightarrow \text{GL}_n(\mathbb{K})$ ;

$$\exp(A) = \sum_{n \geq 0} \frac{1}{n!} A^n = I + A + \frac{1}{2!} A^2 + \frac{1}{3!} A^3 + \dots$$

**Proposition 1.4.** *Let  $A \in M_n(\mathbb{K})$ .*



- (i) For  $u, v \in \mathbb{C}$ ,  $\exp((u + v)A) = \exp(uA) \exp(vA)$ .
- (ii)  $\exp(A)^{-1} = \exp(-A)$ .
- (iii) If  $B \in M_n(\mathbb{K})$  commutes with  $A$ , then  $\exp(A + B) = \exp(A) \exp(B)$ .

We define the it logarithm function by

$$\log : N_{M_n(\mathbb{K})}(I; 1) \longrightarrow M_n(\mathbb{K});$$

$$\log(A) = \sum_{n \geq 1} \frac{(-1)^{n-1}}{n} (A - I)^n$$

**Proposition 1.5.** *The exp and log functions satisfy*

- (i) if  $\|A - I\| < 1$ , then  $\exp(\log(A)) = A$ ;
- (ii) if  $\|\exp(B) - I\| < 1$ , then  $\log(\exp(B)) = B$ .

The functions exp and log are continuous and in fact infinitely differentiable on their domains. By continuity of exp at  $O$ , there is a  $\delta_1 > 0$  such that

$$N_{M_n(\mathbb{K})}(O; r) \subseteq \exp^{-1} N_{GL_n(\mathbb{K})}(I; 1).$$

Since  $\exp N_{M_n(\mathbb{K})}(O; r) \subseteq N_{M_n(\mathbb{K})}(I; e^r - 1)$ , we can take  $\delta_1 = \log 2$ . Hence we have the following result.

**Proposition 1.6.** *The exponential function exp is injective when restricted to the open subset  $N_{M_n(\mathbb{K})}(O; \log 2) \subseteq M_n(\mathbb{K})$ . Hence it is locally a diffeomorphism at  $O$  with the local inverse log.*

## 1.6 Differential Equations in Matrices

**Definition 1.4.** A differentiable curve in  $M_n(\mathbb{K})$  is a function  $\alpha : (a, b) \longrightarrow M_n(\mathbb{K})$  for which the derivative  $\alpha'(t)$  exists for each  $t \in (a, b)$ . Here  $\alpha'(t)$  is defined as an element of  $M_n(\mathbb{K})$  by

$$\alpha'(t) = \lim_{s \rightarrow t} \frac{1}{(s - t)} (\alpha(s) - \alpha(t)),$$

provided that the limit exists.

Consider the first order differential equation  $\alpha'(t) = \alpha(t)A$ , where  $\alpha$  is assumed to be a differentiable curve.

**Theorem 1.5.** *For  $A, C \in M_n(\mathbb{R})$  with  $A$  non-zero, and  $a < 0 < b$ , the differential equation  $\alpha'(t) = \alpha(t)A$ , has a unique solution  $\alpha : (a, b) \longrightarrow M_n(\mathbb{R})$  for which  $\alpha(0) = C$ . Furthermore, if  $C$  is invertible, then so is  $\alpha(t)$  for  $t \in (a, b)$ .*

*Proof.* Consider the function  $\alpha : (a, b) \rightarrow M_n(\mathbb{R})$ ;  $\alpha(t) = \exp(tA)$ . It is differentiable with

$$\alpha'(t) = \sum_{k \geq 1} k \frac{t^{k-1}}{(k-1)!} A^k = A \exp(tA)$$

Hence  $\alpha$  satisfies the differential equation with boundary condition  $\alpha(0) = I$ . Clearly,  $\alpha(t) = C \exp(tA)$  is one solution to  $\alpha(0) = I$ . If  $\beta$  is another such solution, then  $\gamma(t) = \beta(t) \exp(-tA)$  satisfies

$$\begin{aligned} \gamma'(t) &= \beta'(t) \exp(-tA) + \beta(t) \exp(-tA)(-A) \\ &= \beta(t)A \exp(-tA) - \beta(t)A \exp(-tA) \\ &= 0. \end{aligned}$$

Hence  $\gamma$  is a constant function with  $\gamma(t) = \gamma(0) = C$ . Thus  $\beta(t) = C \exp(tA) = \alpha(t)$ . ■

## 1.7 One Parameter Subgroups in Matrix Groups

**Definition 1.5.** A *differentiable curve* in  $G$  is a function  $\gamma : (a, b) \rightarrow G$  for which derivative  $\gamma'(t)$  exists at each  $t \in (a, b)$ .

We usually assume that  $a < 0 < b$  when considering such curves. Now suppose that  $\varepsilon > 0$  or  $\varepsilon = \infty$ .

**Definition 1.6.** A *one-parameter subgroup* in  $G$  is a continuous function  $\gamma : (-\varepsilon, \varepsilon) \rightarrow G$  which is differentiable at 0 and also satisfies  $\gamma(s + t) = \gamma(s)\gamma(t)$  whenever  $s, t, s + t \in (-\varepsilon, \varepsilon)$ . If  $\varepsilon = \infty$ , then  $\gamma : \mathbb{R} \rightarrow G$  is called a *one-parameter subgroup* of  $G$ .

**Proposition 1.7.** Let  $\gamma : (-\varepsilon, \varepsilon) \rightarrow G$  be a one-parameter semigroup. Then for every  $t \in (-\varepsilon, \varepsilon)$ ,  $\gamma$  is differentiable at  $t$ , and

$$\gamma'(t) = \gamma'(0)\gamma(t) = \gamma(t)\gamma'(0).$$

**Proposition 1.8.** Let  $\gamma : (-\varepsilon, \varepsilon) \rightarrow G$  be a one-parameter semigroup. Then there is a unique extension to a one-parameter group  $\tilde{\gamma} : \mathbb{R} \rightarrow G$ .

*Proof.* Let  $t \in \mathbb{R}$ . Then for a large enough natural number  $m$ ,  $t/m \in (-\varepsilon, \varepsilon)$ , hence  $\gamma(t/m), \gamma(t/m)^m \in G$ . Similarly for a second natural number  $n$ ,

$\gamma(t/n), \gamma(t/n)^n \in G$ . Since  $mn \geq m$  and  $mn \geq n$ , we also have  $t/mn \in (-\varepsilon, \varepsilon)$ . Therefore,

$$\begin{aligned}\gamma(t/n)^n &= \gamma(mt/mn) \\ &= \gamma(t/mn)^{mn} \\ &= \gamma(nt/mn)^m \\ &= \gamma(t/m)^m\end{aligned}$$

Thus  $\gamma(t/n)^n = \gamma(t/m)^m$ . This defines a function

$$\tilde{\gamma} : \mathbb{R} \longrightarrow G; \quad \tilde{\gamma}(t) = \gamma(t/n)^n$$

for large  $n$ . It is easy to see that  $\tilde{\gamma}$  is an one-parameter subgroup in  $G$ . ■

**Theorem 1.6.** *Let  $\gamma : \mathbb{R} \longrightarrow G$  be a one-parameter group in  $G$ . Then it has the form  $\gamma(t) = \exp(tA)$  for some  $A \in M_n(\mathbb{K})$ .*

*Proof.* Let  $A = \gamma'(0)$ . Since  $\gamma$  satisfies the differential equation  $\gamma'(t) = \gamma(t)A$ ,  $\gamma(0) = A$ , by Theorem 1.5, this equation has a unique solution  $\gamma(t) = \exp(tA)$ . ■



# Chapter 2

## Algebras and Quaternions

### 2.1 Algebras

**Definition 2.1.** A *finite dimensional algebra*  $A$  is a finite dimensional vector space over  $\mathbb{K}$  which is also an associative and unital ring such that for all  $r, s \in \mathbb{K}$  and  $a, b \in A$ ,  $(ra)(sb) = (rs)(ab)$ .

If every non-zero element  $u \in A$  is invertible, then  $A$  is a *division algebra* over  $\mathbb{K}$ .

**Definition 2.2.** Let  $A$  and  $B$  be two algebras over  $\mathbb{K}$ . A linear transformation  $\varphi : A \rightarrow B$  that is also a ring homomorphism is called a *homomorphism of algebras*.

**Definition 2.3.** An algebra  $A$  is *simple* if it has only one proper two-sided ideal, namely  $(0)$ .

**Proposition 2.1.** Let  $\mathbb{K}$  be a field.

- (i) For a division algebra  $\mathbb{D}$  over  $\mathbb{K}$ ,  $\mathbb{D}$  is simple.
- (ii) For a simple algebra  $A$  over  $\mathbb{K}$ ,  $M_n(A)$  is simple.

**Theorem 2.1** (Wedderburn's Theorem). Let  $A$  be a finite dimensional simple algebra over  $\mathbb{K}$ .

- (i)  $A$  is isomorphic to some  $M_n(\mathbb{D})$ .
- (ii) If  $\mathbb{K}$  is algebraically closed, then  $A$  is isomorphic to some  $M_n(\mathbb{K})$ .
- (i) Every finite dimensional division algebra  $\mathbb{D}$  over  $\mathbb{K}$  has dimension of the form  $\dim_{\mathbb{K}} \mathbb{D} = d^2$  for some  $d \in \mathbb{N}$ .

**Proposition 2.2.** A finite dimensional algebra  $A$  is a product of  $l$  algebras iff there are orthogonal central idempotents  $e_1, \dots, e_l$  such that  $e_1 + \dots + e_l = 1$ .

**Definition 2.4.** An algebra  $A$  is *semi-simple* if it is a product of simple algebras  $A_k$ ,

$$A = A_1 \times \cdots \times A_l$$

**Proposition 2.3.** A finite dimensional algebra  $A$  is semi-simple iff there are indecomposable orthogonal central idempotents  $e_1, \dots, e_l$  such that  $e_1 + \cdots + e_l = 1$ .

## 2.2 Linear Algebra over Division Algebras

Let  $\mathbb{D}$  be a finite dimensional division algebra over a field  $\mathbb{K}$ .

**Definition 2.5.** A (right) vector space  $V$  over  $\mathbb{D}$  is an abelian group with a right scalar multiplication by elements of  $\mathbb{D}$  so that for  $u, v \in V$ ,  $x, y \in \mathbb{D}$ ,

$$\begin{aligned} v(xy) &= (vx)y \\ v(x+y) &= vx + vy \\ (u+v)x &= ux + vx \\ v1 &= v \end{aligned}$$

All obvious notions of linear transformations, subspace, kernels, and images make sense, as do those of spanning sets and linear independence over  $\mathbb{D}$ . The definitions of basis, dimension, and matrices of linear transformations are similar to that of vector spaces over fields.

For a division algebra  $\mathbb{D}$ , each matrix  $A \in M_n(\mathbb{D})$  acts by multiplication on the left of  $\mathbb{D}^n$ . For any subfield  $\mathbb{K}_1 \subseteq \mathbb{D}$  containing  $\mathbb{K}$ ,  $A$  induces a (right) linear transformation over  $\mathbb{K}_1$ ,

$$\mathbb{D}^n \longrightarrow \mathbb{D}^n; \quad x \mapsto Ax.$$

If we choose a basis over  $\mathbb{K}_1$  for  $\mathbb{D}$ ,  $A$  gives rise to a matrix  $\Lambda_A \in M_{nd}(\mathbb{K}_1)$  where  $d = \dim_{\mathbb{K}_1} \mathbb{D}$ .  $A$  is invertible in  $M_n(\mathbb{D})$  iff  $\Lambda_A$  is invertible. Hence to determine invertibility of  $A$ , it suffices to consider  $\det \Lambda_A$ .

The resulting function

$$\text{Rdet}_{\mathbb{K}_1} : M_n(\mathbb{D}) \longrightarrow \mathbb{K}_1; \quad \text{Rdet}_{\mathbb{K}_1}(A) = \det \Lambda_A$$

is called the *reduced determinant* of  $M_n(\mathbb{D})$ , and it is a group homomorphism.

## 2.3 Quaternions

**Proposition 2.4.** If  $A$  is a finite dimensional commutative division algebra over  $\mathbb{R}$ , then either  $A = \mathbb{R}$  or there is an isomorphism of real algebras such that  $A \cong \mathbb{C}$ .

*Proof.* Let  $\alpha \in A$ . Since  $A$  is a finite dimensional real vector spaces, the powers of  $\alpha$  must be linearly dependent, sasy

$$t_0 + t_1\alpha + \cdots + t_m\alpha^m = 0$$

for some  $t_j \in \mathbb{R}$  with  $m \geq 1$  and  $t_m \neq 0$ . We can choose  $m$  to be minimal with these properties. If  $t_0 = 0$ , then

$$t_1 + t_2\alpha + t_3\alpha^2 + \cdots + t_m\alpha^{m-1} = 0$$

, contradicting the minimality of  $m$ ; so  $t_0 \neq 0$ . in fact, the polynomial

$$p(X) = t_0 + t_1X + \cdots + t_mX^m \in \mathbb{R}[X]$$

is irreducible. TO see this, wuppose that  $p(X) = p_1(X)p_2(X)$ ; then as  $A$  is a division algebra, either  $p_1(\alpha) = 0$  or  $p_2(\alpha) = 0$ , contradicting minimality of  $m$  if both  $\deg p_1(X) > 0$  and  $\deg p_2(X) > 0$ .

Consider the real subspace

$$\mathbb{R}(\alpha) = \left\{ \sum_{j=0}^k s_j \alpha^j : s_j \in \mathbb{R} \right\} \subseteq A.$$

Then  $\mathbb{R}(\alpha)$  is easily seen to be a real subalgebra of  $A$  and the elements  $1, \alpha, \dots, \alpha^{m-1}$  form a basis, hence  $\dim_{\mathbb{R}} \mathbb{R}(\alpha) = m$ .

Let  $\gamma \in \mathbb{C}$  be any complex root of the irreducible polynomial

$$t_0 + t_1X + \cdots + t_mX^m \in \mathbb{R}[X].$$

There is a real linear transformation which is actually an injection,

$$\varphi : \mathbb{R}(\alpha) \longrightarrow \mathbb{C}; \quad \varphi \left( \sum_{j=0}^{m-1} s_j \alpha^j \right) = \sum_{j=0}^{m-1} s_j \gamma^j.$$

It is easy to see thaatt this is actually a homomorphism of real algebras. Hence  $\varphi(\mathbb{R}(\alpha)) \subseteq \mathbb{C}$  is a subalgebra. But as  $\dim_{\mathbb{R}} \mathbb{C} = 2$ , this implies that  $m = \dim_{\mathbb{R}} \mathbb{R}(\alpha) \leq 2$ . If  $m = 1$ ,  $\alpha \in \mathbb{R}$ . If  $m = 2$ , then  $\varphi(\mathbb{R}(\alpha)) = \mathbb{C}$ .

So either  $\dim_{\mathbb{R}} A = 1$  and  $A = \mathbb{R}$ , or  $\dim_{\mathbb{R}} A > 1$  and we can choose an  $\alpha \in A$  with  $\mathbb{C} \cong \mathbb{R}(\alpha)$ . This means that we can view  $A$  as a finite dimensional complex algebra. Now for any  $\beta \in A$  there is polynomial

$$q(X) = u_0 + u_1X + \cdots + u_lX^l \in \mathbb{C}[X]$$

with  $l \geq 1$  and  $u_l \neq 0$ . Again choosing  $l$  to be minimal with this property,  $q(X)$  is irreducible. But then sinec  $q(X)$  has a root in  $\mathbb{C}$ ,  $l = 1$  and  $\beta \in \mathbb{C}$ . This shows that  $A = \mathbb{C}$  whenever  $\dim_{\mathbb{R}} A > 1$ . ■

However, the question of what finite dimensional division algebra over  $\mathbb{R}$  exist is less easy to decide. In fact, there is only one other upto an isomorphism, the skew-field of quaternions, denoted  $\mathbb{H}$ . We will construct  $\mathbb{H}$  as a ring of  $2 \times 2$  complex matrices.

Let

$$\mathbb{H} = \left\{ \begin{bmatrix} z & w \\ -\bar{w} & \bar{z} \end{bmatrix} : z, w \in \mathbb{C} \right\} \subseteq M_2(\mathbb{C}).$$

$\mathbb{H}$  is a subring of  $M_2(\mathbb{C})$ , and in fact a reeal subalgebra of  $M_2(\mathbb{C})$ . It also contains a copy of  $\mathbb{C}$ , namely the subalgebra

$$\left\{ \begin{bmatrix} z & 0 \\ 0 & \bar{z} \end{bmatrix} : z \in \mathbb{C} \right\} \subseteq \mathbb{H}.$$

However,  $\mathbb{H}$  is not an algebra over  $\mathbb{C}$ .

An element of  $\mathbb{H}$  is invertible *iff* it is an invertible as a matrix. It is useful to denote a norm in  $\mathbb{H}$  as

$$|H| = \det \begin{bmatrix} z & w \\ -\bar{w} & \bar{z} \end{bmatrix} = |z|^2 + |w|^2.$$

Then  $SU(2) = \{A \in \mathbb{H} : |A| = 1\} \leq \mathbb{H}^\times$ .

As a basis of  $\mathbb{H}$  over  $\mathbb{R}$ , we have the matrices

$$1 = I, \quad i = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}, \quad j = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \quad k = \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}.$$

These satisfy the equations

$$i^2 = j^2 = k^2 = -1, \quad ij = k = -ji, \quad jk = i = -kj, \quad ki = j = -ik.$$

From now on we will write quaternions in the form

$$q = xi + yj + zk + t1$$

$q$  is a pure quaternion *iff*  $t=0$ .

There is a quaternionic analogue of complex conjugation, namely

$$q = xi + yj + zk + t1 \mapsto \bar{q} = -xi - yj - zk + t1$$

This satisfies the properties

$$\begin{aligned}\overline{q_1 + q_2} &= \overline{q_1} + \overline{q_2} \\ \overline{q_1 q_2} &= \overline{q_1} \overline{q_2} \\ \overline{q} = q &\iff q \text{ is a real quaternion;} \\ \overline{q} = -q &\iff q \text{ is a pure quaternion.}\end{aligned}$$

The inverse of a non-zero quaternion can be written as

$$q^{-1} = \frac{\overline{q}}{q\overline{q}}$$

In terms of the matrix description of  $\mathbb{H}$ , quaternionic conjugation is given by the hermitian conjugation,

$$\begin{bmatrix} z & w \\ -\bar{w} & \bar{z} \end{bmatrix} \mapsto \begin{bmatrix} \bar{z} & -w \\ \bar{w} & z \end{bmatrix}.$$

## 2.4 Quaternionic Matrix Groups

The above norm — on  $\mathbb{H}$  extends to a norm on  $\mathbb{H}^n$ , viewed as a right  $\mathbb{H}$ -vector space. We can define a quaternionic inner product on  $\mathbb{H}$  by

$$x \cdot y = x^* y = \sum_{r=1}^n \bar{x}_r y_r,$$

where we define the *quaternionic conjugate* of a vector by

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}^* = [\bar{x}_1 \quad \bar{x}_2 \quad \cdots \quad \bar{x}_n].$$

Similarly, for any matrix  $[a_{ij}]$  over  $\mathbb{H}$  we can define  $[a_{ij}]^* = [\bar{a}_{ji}]$ .

The length of  $x \in \mathbb{H}^n$  is defined to be  $|x| = \sqrt{x * x}$ .

## 2.5 Automorphism Groups of Algebras





# Chapter 3

## Lie Algebras and Tangent Spaces

### 3.1 Lie Algebras

**Definition 3.1.** A Lie algebra over  $\mathbb{K}$  consists of a vector space  $\mathfrak{a}$  over a field  $\mathbb{K}$ , together with a bilinear map  $[\cdot, \cdot] : \mathfrak{a} \times \mathfrak{a} \longrightarrow \mathfrak{a}$  called the *Lie bracket*, such that for  $x, y, z \in \mathfrak{a}$ ,

$$[x, y] = -[y, x],$$
$$\text{and } [x, [y, z]] + [y, [x, z]] + [z, [x, y]] = 0$$

Given two matrices  $A, B \in M_n(\mathbb{K})$ , their *commutator* is the *Lie Bracket* is the matrix  $[A, B] = AB - BA$ . This defines a bilinear function

$$[\cdot, \cdot] : M_n(\mathbb{K}) \times M_n(\mathbb{K}) \longrightarrow M_n(\mathbb{K}); \quad (A, B) \mapsto [A, B]$$

which easily satisfies the definition. Hence  $M_n(\mathbb{K})$  is a Lie algebra over  $\mathbb{K}$  with the commutator as its Lie bracket.

If all brackets in a Lie algebra are trivial, it is called *abelian*. We further develop some of the basic algebraic notions of Lie algebra theory, analogous to the basic theory of groups and rings.

**Definition 3.2.** If  $\mathfrak{a}$  is a Lie algebra over  $\mathbb{K}$ , then a subspace  $\mathfrak{b} \subseteq \mathfrak{a}$  is a *Lie subalgebra* of  $\mathfrak{a}$  if it is closed under the Lie bracket, that is, whenever  $x, y \in \mathfrak{b}$ ,  $[x, y] \in \mathfrak{b}$ ; we write  $\mathfrak{b} \leq \mathfrak{a}$ .

$\mathfrak{a}$  is a Lie subalgebra of itself, and so is  $\{0\}$ .

**Definition 3.3.** If  $\mathfrak{a}$  is a Lie algebra over  $\mathbb{K}$ , then a vector subspace  $\mathfrak{n} \subseteq \mathfrak{a}$  is a *Lie ideal* of  $\mathfrak{a}$ , given  $[z, x] \in \mathfrak{n}$  for all  $z \in \mathfrak{n}$  and  $x \in \mathfrak{a}$ ; we then write  $\mathfrak{n} \triangleleft \mathfrak{a}$ .

A Lie ideal is *proper* if  $\mathfrak{n} \neq \mathfrak{a}$ , and is *non-trivial* if  $\mathfrak{n} \neq 0$ .

**Definition 3.4.** The *centre* of  $\mathfrak{a}$  is the subset

$$z(\mathfrak{a}) = \{z \in \mathfrak{a} : \forall x \in \mathfrak{a}, [z, x] = 0\}$$

**Definition 3.5.** The *derived subalgebra*  $\mathfrak{a}' \leq \mathfrak{a}$  is the vector subspace spanned by all the brackets  $[x, y]$ ,  $x, y \in \mathfrak{a}$ .

**Proposition 3.1.**  $z(\mathfrak{a})$  and  $\mathfrak{a}'$  are Lie ideals of  $\mathfrak{a}$ .

**Definition 3.6.** A Lie algebra is *simple* if every proper ideal of the Lie algebra is trivial.

**Definition 3.7.** Let  $\mathfrak{g}, \mathfrak{h}$  be Lie algebras. A linear transformation  $\phi : \mathfrak{g} \rightarrow \mathfrak{h}$  is a *homomorphism of Lie algebras* if for all  $x, y \in \mathfrak{g}$ ,  $\phi([x, y]) = [\phi(x), \phi(y)]$ .

**Proposition 3.2.** Let  $\phi : \mathfrak{a} \rightarrow \mathfrak{b}$  be a homomorphism of Lie algebras. Then  $\ker \phi \triangleleft \mathfrak{a}$  and there is an isomorphism  $\text{im } \phi \cong \mathfrak{a}/\ker \phi$ .

## 3.2 Curves and Tangent Spaces

Throughout this section, let  $G \subseteq GL_n(\mathbb{K})$  be a matrix group.

**Definition 3.8.** The *tangent space* to  $G$  at  $U \in G$  is

$$T_U G = \{\gamma'(0) \in M_n(\mathbb{K}) : \gamma \text{ is a differentiable curve in } G \text{ with } \gamma(0) = U\}$$

**Proposition 3.3.**  $T_U G$  is a real vector subspace of  $M_n(\mathbb{K})$ .

*Proof.* Suppose  $\alpha, \beta$  are differentiable curves in  $G$  for which  $\alpha(0) = \beta(0) = U$ . Then

$$\gamma : \text{dom } \alpha \cap \text{dom } \beta \rightarrow G; \quad \gamma(t) = \alpha(t)U^{-1}\beta(t)$$

is also a differentiable curve in  $G$  with  $\gamma(0) = U$ . Now,

$$\gamma'(t) = \alpha'(t)U^{-1}\beta(t) + \alpha(t)U^{-1}\beta'(t)$$

Hence  $\gamma'(0) = \alpha'(0) + \beta'(0)$ , which shows that  $T_U G$  is closed under addition.

Similarly, if  $r \in \mathbb{R}$  and  $\alpha$  is a differentiable curve in  $G$  with  $\alpha(0) = U$ , then  $\eta(t) = \alpha(rt)$  defines another such curve. Since  $\eta'(0) = r\alpha'(0)$ , we see that  $T_U G$  is closed under real scalar multiplication. ■

**Definition 3.9.** The *dimension* of a real matrix group  $G$  is  $\dim G = \dim_{\mathbb{R}} T_I G$ . If  $G$  is complex, then its *complex dimension* is  $\dim_{\mathbb{C}} G = \dim_{\mathbb{C}} T_I G$ .

We denote  $T_I G = \mathfrak{g}$ . In fact,  $\mathfrak{g}$  has the algebraic structure of a real Lie algebra.

**Theorem 3.1.** *If  $G$  is a matrix subgroup, then  $\mathfrak{g} = T_1 G$  is a real Lie subalgebra of  $M_n(\mathbb{K})$ . If  $G$  is complex matrix subgroup and  $\mathfrak{g}$  is a subspace of  $M_n(\mathbb{C})$ , then  $\mathfrak{g}$  is a complex Lie subalgebra.*

*Proof.* We will show that for differentiable curves  $\alpha, \beta$  in  $G$  which satisfy  $\alpha(0) = \beta(0) = I_n$ , there is another such curve  $\gamma$  with  $\gamma'(0) = [\alpha'(0), \beta'(0)]$ .

Consider the function

$$F : \text{dom } \alpha \times \text{dom } \beta \longrightarrow G; \quad F(s, t) = \alpha(s)\beta(t)\alpha(s)^{-1}.$$

This curve is clearly continuous and differentiable with respect to each of the variables  $s$  and  $t$ . For each  $s \in \text{dom } \alpha$ , the function  $F(s, \cdot) : \text{dom } \beta \longrightarrow G$  is a differentiable curve in  $G$  with  $F(s, 0) = I_n$ . On differentiating we find

$$\left. \frac{d}{dt} \right|_{t=0} F(s, t) = \alpha(s)\beta'(0)\alpha(s)^{-1},$$

and so  $\alpha(s)\beta'(0)\alpha(s)^{-1} \in \mathfrak{g}$ . Since  $\mathfrak{g}$  is a closed subspace of  $M_n(\mathbb{K})$ , whenever this limit exists, we also have

$$\lim_{s \rightarrow 0} \frac{1}{s} (\alpha(s)\beta'(0)\alpha(s)^{-1} - \beta'(0)) \in \mathfrak{g}.$$

Using the matrix rule for differentiating an inverse, we have

$$\begin{aligned} \lim_{s \rightarrow 0} \frac{1}{s} (\alpha(s)\beta'(0)\alpha(s)^{-1} - \beta'(0)) &= \left. \frac{d}{ds} \right|_{s=0} \alpha(s)\beta'(0)\alpha(s)^{-1} \\ &= \alpha'(0)\beta'(0)\alpha(0) - \alpha(0)\beta'(0)\alpha(0)^{-1}\alpha'(0)\alpha(0)^{-1} \\ &= \alpha'(0)\beta'(0) - \beta'(0)\alpha'(0) \\ &= [\alpha'(0), \beta'(0)] \end{aligned}$$

Hence  $[\alpha'(0), \beta'(0)]$  must be the form  $\gamma'(0)$  for some differentiable curve. ■

So for each matrix group  $G$  there is a Lie algebra  $\mathfrak{g} = T_1 G$ . A suitable type of homomorphism  $G \longrightarrow H$  between matrix groups gives rise to a linear transformation  $\mathfrak{g} \longrightarrow \mathfrak{h}$  which is a homomorphism of Lie algebras.

**Definition 3.10.** Let  $G, H$  be matrix groups, and  $\varphi : G \longrightarrow H$  be a continuous map. Then  $\varphi$  is said to be a *differentiable map* if, for every differentiable curve  $\gamma : (a, b) \longrightarrow G$ , the curve  $\varphi \circ \gamma : (a, b) \longrightarrow H$  is differentiable and has derivative  $(\varphi \circ \gamma)'(t) = \frac{d}{dt} \varphi(\gamma(t))$ ; and if two differentiable curves satisfy  $\alpha(0) = \beta(0)$ , and  $\alpha'(0) = \beta'(0)$ , then  $(\varphi \circ \alpha)'(0) = (\varphi \circ \beta)'(0)$ .

A differentiable map which is also a group homomorphism is called a *differential homomorphism*. A continuous homomorphism of matrix groups that is also differentiable is called a *Lie homomorphism*.

**Proposition 3.4.** *Let  $G, H, K$  be matrix groups with  $\varphi : G \rightarrow H$  and  $\theta : H \rightarrow K$  differential homomorphisms. Then,*

(i) *for each  $A \in G$ , there is a real linear transformation  $d\varphi_A : T_A G \rightarrow T_{\varphi(A)} H$  given by  $d\varphi_A(\gamma'(0)) = (\varphi \circ \gamma)'(0)$ , for every differentiable curve  $\gamma : (a, b) \rightarrow G$  with  $\gamma(0) = A$ ;*

(ii) *we have  $d\theta_{\varphi(A)} \circ d\varphi_A = d(\theta \circ \varphi)_A$ ;*

(iii) *for the identity map  $\text{Id}_G$ ,  $d\text{Id}_G = \text{Id}_{T_A G}$ .*

**Theorem 3.2.** *Let  $G, H$  be matrix groups and  $\varphi : G \rightarrow H$  be a differential homomorphism. Then the derivative  $d\varphi : \mathfrak{g} \rightarrow \mathfrak{h}$  is a homomorphism of Lie algebras.*

*Proof.* Following the ideas and notations from the proof of theorem 3.1, for differentiable curves  $\alpha, \beta$  in  $G$  with  $\alpha(0) = \beta(0) = I$ , we can use the composite function  $\varphi \circ F$  given by

$$\varphi \circ F(s, t) = \varphi(F(s, t)) = \varphi(\alpha(s)\beta(s)\alpha(s)^{-1})$$

to deduce that

$$d\varphi([\alpha'(0), \beta'(0)]) = [d\varphi(\alpha'(0)), d\varphi(\beta'(0))].$$

■

### 3.3 Lie Algebras of Some Matrix Groups

Here we determine the Lie algebras of some important families of real and complex matrix groups.

1. General and Special Linear Groups
2. Upper Triangular and Unipotent Groups
3. Affine Groups
4. Orthogonal and Special Orthogonal Groups
5. Unitary and Special Unitary Groups
6. Quaternionic Symplectic Group

### 3.4 $SO(3)$ and $SU(2)$

The lie algebras  $\mathfrak{so}(3)$  and  $\mathfrak{su}(2)$  are both 3-dimensional vector spaces, for example with the following bases:

$$\mathfrak{so}(3): \quad P = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad Q = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \quad R = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix},$$

$$\mathfrak{su}(2): \quad H = \frac{1}{2} \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}, \quad E = \frac{1}{2} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \quad F = \frac{1}{2} \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}.$$

The non trivial Lie brackets among these are,

$$[P, Q] = R, \quad [Q, R] = P, \quad [R, P] = Q,$$

$$[H, E] = F, \quad [E, F] = H, \quad [F, H] = E.$$

This implies that the real linear isomorphism  $\varphi : \mathfrak{su}(2) \longrightarrow \mathfrak{so}(3)$ ;

$$\varphi(xH + yE + zF) = xP + yQ + zR \quad (x, y, z) \in \mathbb{R}$$

satisfies  $\varphi([U, V]) = [\varphi(U), \varphi(V)]$ , so is an isomorphism of real Lie Algebras. Thus these Lie algebras look the same algebraically. This suggests that there might be a close relationships between the groups themselves. Notice also that both  $\mathfrak{su}(2)$  and  $\mathfrak{so}(3)$  are isomorphic to  $\mathbb{R}^3$  with the vector cross product as its Lie bracket.



# **Chapter 4**

## **Lie Groups**

**4.1 Smooth Manifolds**

**4.2 Tangent Spaces and Derivatives**

**4.3 Lie Groups**

**4.4 Some Examples of Lie Groups**

**4.5 Some Useful Formulae in Matrix Groups**

**4.6 Matrix Groups are Lie Groups**

**4.7 Not All Matrix Groups are Matrix Groups**

**▲▼▲**

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